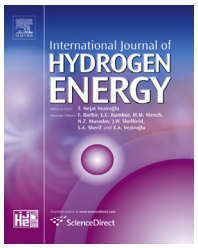
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Substitutional V–Pd alloys for the membranes permeable to hydrogen: Hydrogen solubility at 150–400 °C

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ARTICLE INFO

Article history:
Received 30 April 2014
Received in revised form
19 September 2014
Accepted 22 September 2014
Available online 14 October 2014

Keywords:
Hydrogen separation
Vanadium alloys
Hydrogen solubility
Composite membranes
Palladium

ABSTRACT

The development of non-palladium membrane for separation of hydrogen from gas mixtures is one of critical challenges of hydrogen energy. Vanadium based materials are most promising for such membranes. The alloying of pure vanadium is crucially important for reduction of hydrogen solubility to an optimal value. Solution of hydrogen in substitutional V-xPd alloys (x = 5, 7.3, 9.7, 12.3, 18.8 at%) was investigated. The pressure–composition-isotherms were obtained in the range of pressure (10–10^6) Pa, temperature (150–400) °C and concentration of hydrogen, H/M, from $4 \cdot 10^{-4}$ to 0.6. The alloying of vanadium with palladium was found to reduce the hydrogen solubility substantially greater than the alloying with other elements, e.g. by Ni and Cr. The hydrogen absorption in the V–Pd alloys obeyed Siverts' law including the range of undiluted solution with hydrogen concentration H/M > 0.1. The reduction in the hydrogen solubility due to the alloying of V with Pd was caused mainly by increase in the enthalpy of solution at nearly constant entropy factor. Changes in the gross electronic structure of metal are most probably responsible for the effects of alloying on the hydrogen solubility in the substitutional V–Pd alloys.
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Introduction

There are a number of reasons why the hydrogen solubility in V alloys and in particularly in the V–Pd ones is of interest to basic researches and for practical applications.

First of all the V based alloys are promising materials for selective membranes for the hydrogen separation from gas mixtures including the extraction of ultrapure hydrogen for fuel cells from the products of hydrocarbon reforming [1–11]. This is because the speed of hydrogen transport through the lattice of group 5 metals is orders-of-magnitude higher than through any other metallic lattice including the palladium one [1,12–14]. Notice that the speed of transcrystalline hydrogen transport through vanadium is greatest among the group 5 metals [1,2,14,15].

The problem is that the hydrogen solubility in group 5 metals is too high [14,15]. As a result the concentration of

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http://dx.doi.org/10.1016/j.ijhydene.2014.09.122
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dissolved hydrogen at pressures of practical interest may exceed a limit admissible from the viewpoint of mechanical stability of material. In principle, the concentration of absorbed hydrogen in group 5 metals may be reduced by increase of temperature (hydrogen absorption in these metals is strongly exothermic [14,15]). However, if the membrane is made of group 5 metal, its surfaces should be covered with a layer of Pd or Pd alloy [1–11] whose thermal stability is limited. Consequently membrane temperature typically should not exceed 400 °C [1,2,4,7,10]. Although the hydrogen solubility in vanadium is the lowest among the group 5 metals [14,15], the equilibrium concentration of dissolved hydrogen, C, at 400 °C exceeds the value of H/M = 0.3 at pressure of hydrogen, P, higher than 0.1 MPa and reaches H/M ≈ 0.5 at P = 0.5 MPa [15]. Such a high concentration of absorbed hydrogen at the pressures of interest for the practical applications gives rise to several problems.

(1) The great hydrogen dilatation may result in failure of connection tightness between the membranes and the structural parts.
(2) The hydrogen dilatation may lead to a destructive mechanical stress caused by the gradient of hydrogen concentration across the membrane.
(3) It is likely that vanadium gets brittle at so high concentration of hydrogen even at relatively high temperature (e.g. 400 °C). According to Yukawa et al., the ductile-to-brittle transition occurs for pure vanadium at 400 °C when H/M = 0.22 (i.e. at P > 0.05 MPa) [3].

To improve properties of membrane materials based on V it would be desirable to decrease concentration of absorbed hydrogen at the typical operation conditions. Such reduced/optimized hydrogen solubility may be inherent to some vanadium alloys. If these alloys are disordered solutions of alloying elements in vanadium lattice, hydrogen diffusivity is anticipated to be high as it is typical for bcc lattice.

A number of such binary and ternary alloys were tested [5–10] and some of them demonstrated rather good properties with regards to the hydrogen solubility and permeability. In particular it should be mentioned the study of hydrogen solubility in substitutional Nb–Pd alloys specially focused on the membrane applications [11]. The main problem of all such alloys is their insufficient ductility resulting in difficulties with the fabrication of fine rolled products (sheets or tubes). It was reported that one of promising alloys is V-10 at% Pd which demonstrated an optimal permeability to hydrogen and relatively good ductility [7]. However there are no data about the hydrogen absorption in this alloy as well as in other V alloys with Pd.

There is also another reason why the hydrogen solubility in V–Pd alloys is specifically important for the operation of composite membranes made of V based alloys with any allowing elements (not only with Pd). All such membranes are plated with Pd (or Pd based alloys, e.g. Pd–Ag or Pd–Cu) [1–11], and the formation of V–Pd alloys due to the interdiffusion mixing between the bulk material and Pd coating is considered as a main cause of membrane degradation [1,2,7,16]. Therefore the investigation of hydrogen solubility in the V–Pd alloys with different proportion of components is necessary also for clarifying mechanisms of the membrane degradation. Currently the data on hydrogen solubility in the substitutional Pd alloys with V as alloying element (up to 10 at% V) are available [17]. The present article provides the data for substitutional V alloys with Pd as alloying element (from 5 to 18.8 at% Pd).

Finally hydrogen behavior in the group 5 metals and their alloys is a traditional subject of basic studies [15,17–20]. In particular the effects of different allowing elements (Cr [18,20], Ti, Nb [18], Ni [6]) on the solution of hydrogen in substitutional V alloys were investigated and the mechanisms responsible for changes in the hydrogen solubility were discussed [17–20].

In this work the hydrogen solubility was measured in substitutional V–Pd alloys with the content of Pd 5, 7.3, 9.7, 12.3, 18.8 at% as well as in pure V. The pressure–composition-isotherm (*PCT*) data were obtained in the range of pressure ($10–10^6$) Pa, temperature (150–400) °C and concentration of hydrogen, H/M, from $4\cdot10^{-4}$ to 0.6. Only disordered solution of hydrogen (α phase) was the subject of present study, since only minimum operation temperature (150 °C) approached the critical temperature, T_k, of α–β phase transition for pure V while other operation temperatures substantially exceeded T_k. The solubility in the V–Pd alloys is compared with that for other substitutional V alloys as well as for pure V. The mechanisms responsible for the effects of alloying of V with Pd on the hydrogen solubility are discussed in the context of current ideas about the hydrogen absorption in the substitutional alloys.

Experimental

Sample preparation/characterization

The 100 μm-thick foils made of V–Pd alloys with 5, 7.3, 9.7, 12.3 and 18.8 at% Pd were supplied by «MEVODENA» OJSC. The alloys were prepared from raw materials with the purity of 99.96 wt% for V and 99.9 wt% for Pd. Ribbon-shaped specimens of 0.8 cm in width and 10 cm in length were cut from these foils. Both surfaces of each specimen were electrolessly plated with 2 μm layer of Pd to promote dissociative/associative processes at the gas–metal interface [2,4]. The nominal composition of each alloy and its homogeneity were confirmed by analysis of Pd/V concentration distribution over (across and along) each specimen by means of energy-dispersive X-ray microanalysis (EDX).

The PdV3 phase can be formed when content of Pd in the alloy exceeds 15 at% [21] and therefore PdV3 phase could present in the specimen with maximum concentration of Pd (18.8 at%). Still XRD analysis (with Rigaku Ultima IV multipurpose X-ray diffractometer) detected only peaks assignable to V (though slightly shifted). Thus all the investigated alloys consisted of the single bcc phase of disordered solid solution of Pd in V.

Specific electrical resistance of Pd–V alloys, ρ_{V-Pd}, and of pure V, ρ_V, was measured. The ratio ρ_{V-Pd}/ρ_V at room temperature is presented in Fig. 1 versus the concentration of Pd. The monotonic rise of ρ_{V-Pd}/ρ_V with concentration of Pd is one more evidence of single-phase structure and homogeneity of the alloys.

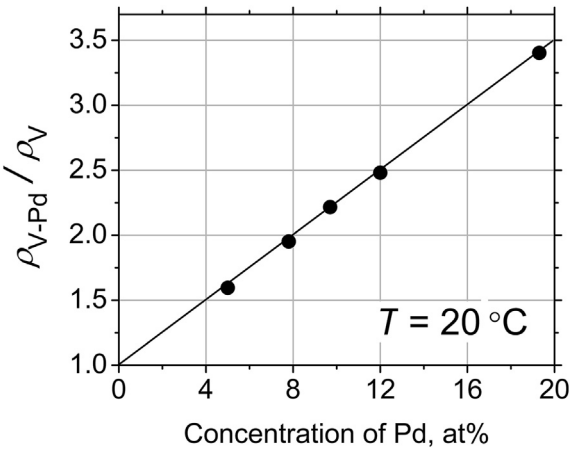


Fig. 1 – The ratio of specific electrical resistance of V–Pd alloys and of pure V *vs* concentration of Pd in the alloy.

PCT measurements

The equilibrium *PCT* data at temperatures between 150 °C and 400 °C and pressures between 10 Pa and 10^6 Pa were determined using a Sieverts-type apparatus. The specimen of V–Pd alloy with given content of Pd (0, 5, 7.3, 9.7, 12.3 and 18.8 at%) was put into apparatus and the pressure–composition (*P–C*) dependence was investigated first at highest temperature (400 °C) and then successively at 350 °C, 250 °C, 200 °C and 150 °C.

In order to determine the *P–C* dependence at a given temperature, the hydrogen pressure was step-by-step increased from the lowest to the maximum value. After reaching an equilibrium at the given pressure, hydrogen was evacuated and the sample was kept under vacuum at the given temperature until the residual concentration of absorbed hydrogen drops to negligible level (due to the palladium coating it occurs quickly enough, e.g. for ≈15 min at 200 °C). Then, hydrogen was injected again and the equilibrium was reached at the next (higher) pressure. Upon completion of the measurement of *P–C* dependence at given temperature, the sample was thoroughly degassed at 400 °C and then was extracted for the inspection. Then the *P–C* measurements were continued at the next (lower) temperature. After finishing the *PCT* measurements with the given sample, the next specimen with another content of Pd in the V–Pd alloy was investigated in the same manner.

Since the equilibrium content of hydrogen was measured for the whole specimen including the Pd coating, we subtracted the portion of H absorbed in Pd when the hydrogen concentration in the alloys was computed from experimental data (we used reference data for solubility of hydrogen in Pd for this calculation [14]). This correction was not large: a few percent for the alloy with the maximum content of Pd (V-18.8Pd) and it is the smaller the lower concentration of Pd in the alloy.

Observation of sample integrity

Hydrogen embrittlement is one of the serious problems in the application of metallic membranes for hydrogen separation. That is why we studied the capability of the ribbon specimens of V–Pd alloys of different composition to keep their integrity under the hydrogen loading at different temperatures. We started the *P–C* measurements at highest *T* (400 °C) where the risk of specimen destruction was minimum. Then we continued the experiment reducing temperature step-by-step (with step of 50 °C) and extracting the specimen after one or two steps from the apparatus for its visual inspection and photography. The samples were thoroughly degassed before their extraction as well as before their cooling down to temperature lower than temperature of last absorption experiment (by heating in vacuum at 400 °C).

Experimental results

Absorption isotherms (400, 350, 300, 250, 200 and 150 °C) obtained for the V-*x*Pd-H_2 system are shown in Fig. 2 for samples ranging in palladium content, *x*, from 5 to 18.8 at% as well as for pure vanadium ($x = 0$) and palladium. The hydrogen concentration, *C*, is defined as the atom ratio, $C \equiv H/M$, where $M = V + Pd$. The data for pure vanadium presents both the results of our experiments (for 300, 350 and 400 °C) and the literature data [14,15]. Thus one can compare the results obtained with the same method for the samples of V–Pd alloys and for referenced samples of pure vanadium. On the other hand, the comparison of our data for pure vanadium with the reference data shows an accuracy of our measurements in different ranges of *P*, *C*, *T* (Fig. 2). This accuracy was the better the higher pressure and the lower solubility of hydrogen in the given material. Due to the latter the accuracy of the absorption measurements for the alloys were in general higher than that for pure vanadium (especially, at lower *T* and *P*).

As follows from the presented data, palladium as an alloying element greatly reduces the ability of vanadium to absorb hydrogen over the whole range of *P* and *T* investigated (even when Pd concentration is as low as 5 at%).

Siverts' law: extending to the range of undiluted solutions

The solubility of hydrogen in pure V observed in the present experiment was close to the reference data [14,15] (Fig. 2). According to these data, until the concentration of solute hydrogen, *C*, remains lower than $H/M \approx 0.05$, the *P–C* curves for pure V well obey Siverts' law

$$C = S_V\sqrt{P} \quad (1)$$

where S_V is Siverts' constant for pure V. When *C* exceeds $H/M \approx 0.05$, the *P–C* curves for pure V deviates from Siverts' law behavior. First their slope reduces with further rise of *C* (due to an attraction between absorbed H atoms [14,15,19,22]). The lower temperature the more pronounced is this reduction in the slope. When temperature decreases lower 150 °C a horizontal plateau appears on *P–C* curves for V due to formation of hydride phase coexisting with the phase of hydrogen solution [14,15] (this trend is demonstrated in Fig. 2(f)). At even higher concentration of absorbed hydrogen ($H/M > 0.4$, Fig. 2) the slope of *P–C* curves for pure V sharply increases because of approaching the maximum hydrogen content, $(H/M)_{max}$,

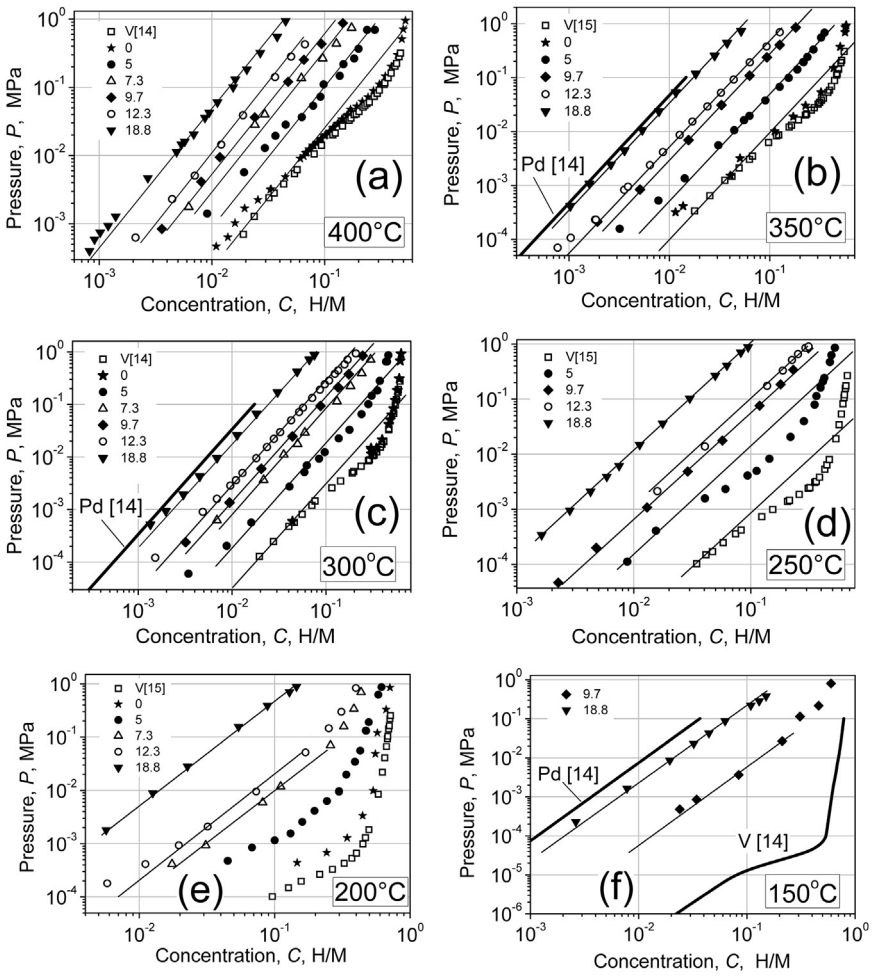


Fig. 2 – Pressure–composition isotherms for hydrogen absorption in substitutional V-xPd alloys with different contents of Pd (x = 0, 5, 7.3, 9.7, 12.3 and 18.8 at%). Literature data for pure V [14,15] are presented along with the experimental data (x = 0) obtained in the present work. Literature data for pure Pd are also presented. Straight lines correspond to Siverts' law (H/M ∞ $\sqrt{P}$). (a): 400 °C, (b): 350 °C, (c): 300 °C, (d): 250 °C, (e): 200 °C, (f): 150 °C.

which would be possible without change in mechanism of solution [22].

According to presented P–C data (Fig. 2), absorption of hydrogen in the V–Pd alloys also obeyed Siverts' law in a definite region of C

$$C = S_{V-xPd}\sqrt{P} \quad (2)$$

where S_{V-xPd} is the Siverts' constant for V-xPd alloy and x is the concentration of Pd in the alloy (in at%). A peculiar feature of solution of hydrogen in the V–Pd alloys is that Siverts' law remained valid up to a substantially higher concentration of absorbed hydrogen than in the case of hydrogen absorption in pure V. As demonstrated in Fig. 2, the Siverts' law is obeyed up to the concentration H/M, which is the higher the greater the content of Pd in the alloy. Moreover, for the higher concentrations of Pd (x ≥ 9.7 at%), the Siverts' law extends far beyond the range of C where the solution can be considered as dilute (up to H/M ≈ 0.3, Fig. 2(b)–(d)). The similar effect of allowing of V with Ti, Nb and Cr on the hydrogen solution behavior was reported by Peterson and Nelson [18].

Temperature dependence of S_{V-xPd}

Arrhenius plot of hydrogen solution constants for the investigated V–Pd alloys, $S_{V\text{-}xPd}$, (x = 5, 7.3, 9.7, 12.3 and 18.8 at%) as well as for pure V and Pd is shown in Fig. 3. The plot was constructed from the P–C isotherms presented in Fig. 2 for the range of C and P where Siverts' law is valid. The Arrhenius graphs for $S_{V\text{-}xPd}$ can be approximated by straight lines without any breaks in their slope over the whole temperature range (150 °C–400 °C) for all the alloys investigated. Being extrapolated to 1/T = 0, these lines intersect the solubility axis close to each other and to the cross point of similar line for pure vanadium (Fig. 3) (while rather far from that for pure Pd). That means that the pre-exponential (entropy) factor of Siverts' constant, S^*_{V-xPd}, for V–Pd alloys is close to that for V, S^*_V.[1] Therefore, the difference in the hydrogen solution

[1] Notice that entropy factors for Siverts' constant are rather close for different bcc metals, e.g. for V, Nb, Ta, while all of them as a rule are much smaller than those for fcc metals including Pd [14].

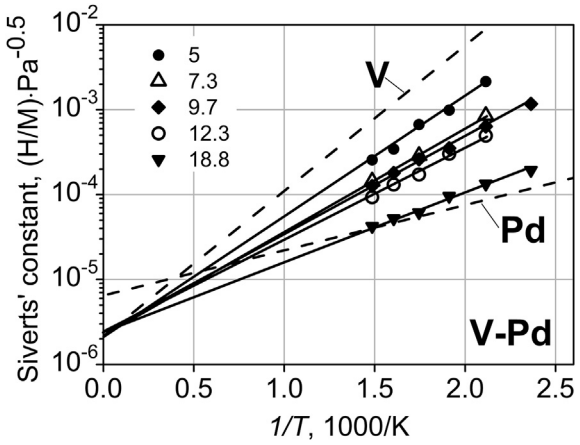


Fig. 3 – Temperature dependence of Siverts' constant for V-xPd alloys as well as for pure V and Pd [14]. The concentration of Pd in the alloys, x, is indicated (in at%).

constants for V and V-xPd alloys is caused practically by only difference in the enthalpy of solution. The similar result was reported by Lynch et al. for V–Cr alloys [20].

Thus with the accuracy of our experimental data we can take $S^*_{V-xPd} = S^*_V$ and present the hydrogen solution constants for all the investigated alloys as

$$S_{V-xPd} = S^*_V \exp\frac{\Delta H_{V-xPd}}{RT} \quad (3)$$

where ΔH_{V-xPd} is the molar enthalpy of solution of hydrogen in the V-xPd alloys. According to the above, Eq. (3) may be valid for V-xPd alloys in the range of C that is far beyond the range of dilute solution.

Dependence of ΔH_{V-xPd} on the concentration of Pd found from the Arrhenius plot is shown in Fig. 4. The similar dependence for substitutional V-xCr alloys plotted from the data by Peterson and Nelson [18] are given for comparison. It seems to be noteworthy that the dependence of enthalpy of solution of hydrogen on the concentration of alloying element is nearly linear for both elements Pd and Cr though the effect of Pd turned out much greater than that of Cr.

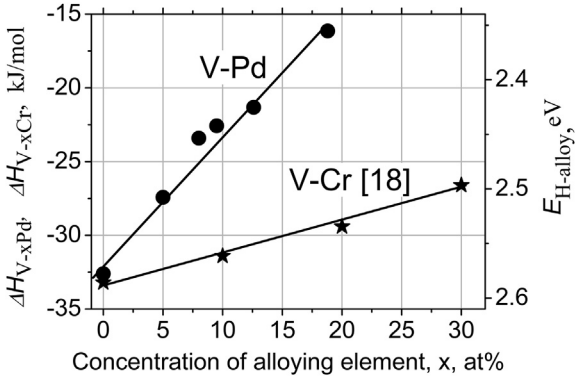


Fig. 4 – The enthalpy of solution of hydrogen (left axis) and the binding energy of H atom with metal lattice (right axis) in substitutional alloys V-xPd and V-xCr [18] as a function of content of alloying element, x.

In Fig. 5 the temperature dependence of Siverts' constant for the V–Pd alloy with 9.5 at% Pd is compared with that for the V–Ni [6] and V–Cr [18] alloys with 10 at% of alloying element. Fig. 5 demonstrates that effect of Pd doping on the hydrogen solubility in vanadium is greater than that of Ni and Cr. Furthermore, compared with Ni and Cr, doping of V with Pd leads to a much more pronounced weakening in the temperature dependence of hydrogen solubility.

The dependence on the concentration of Pd

Ratio of Siverts' constants for V-xPd alloys and for pure V, S_{V-xPd}/S_V, is presented in Fig. 6 as function of concentration of Pd, x. The data in the range of $0 \le x \le 18.8$ at% are found from the P–C isotherms presented in Fig. 2. The dependence of S_{V-xPd}/S_V on x can be well approximated by an exponent in correspondence with the linear dependence of ΔH_{V-xPd} on x (Fig. 4) and nearly constancy of S^*_{V-xPd} (Fig. 3). Note, in this range of x the V-xPd alloys are disordered solid solutions of Pd in bcc lattice of V.

S_{V-xPd}/S_V for the range of $90 \le x \le 100$ at% (Fig. 6) was computed from the data by Sakamoto et al. [17] on the hydrogen solubility in Pd–V alloys. The alloys in this range of x are disordered solid solutions of V in the fcc lattice of Pd. Interestingly, both the substitution of V by Pd and the replacement of Pd by V (in the bcc and fcc lattices respectively) greatly suppress the hydrogen solubility.

Capabilities of different alloying elements (Pd, Cr and Ni) to reduce hydrogen solubility in the substitutional alloys of V (in solid bcc solutions) are compared in Fig. 7, where the ratio of Siverts' constant for the vanadium alloys, S_{alloy}, and for pure vanadium, S_V, is presented. Though Pd itself is the stronger absorber of hydrogen than Cr and Ni, it suppresses the hydrogen solubility in the binary substitutional alloys of V much stronger than Cr, Ni, as well as than W [3]. Notice that the effect of Ti, Nb and Cr on the hydrogen solubility in substitutional alloys of V was found to be in accordance with the affinity of these metals to hydrogen [18]. The effect of Pd on hydrogen solubility in substitutional alloys of V seems to be greatest ever observed with any other alloying elements. The

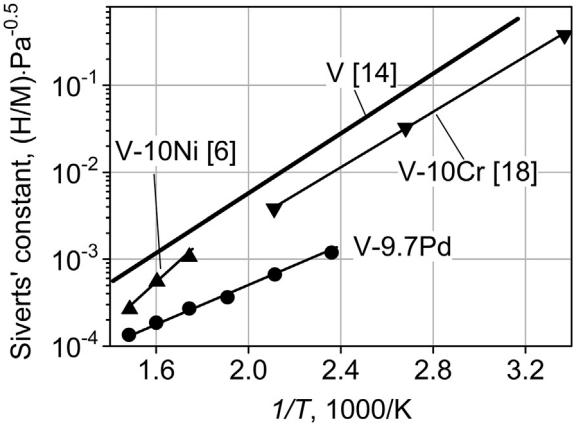


Fig. 5 – Comparison of temperature dependences of Siverts' constants for different substitutional binary alloys of vanadium: V-9.7 Pd, V-10 Cr [18] and V-10 Ni [6]) as well as for pure V [14].

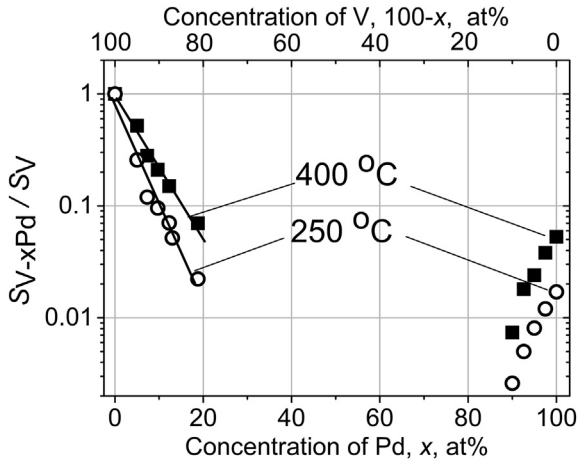


Fig. 6 – Ratio of Siverts' constants for V-xPd alloys and for pure V, S_{V-xPd}/S_V, as a function of concentration of Pd, x. The data in the range of $0 \le x \le 18.8$ at% were found from the *PCT* plot presented in Fig. 2. The data in the range of $90 \le x \le 100$ at% were computed from the data by Sakamoto et al. [17].

same was found for substitutional alloys of Nb [23]: reduction in the hydrogen solubility was stronger due to the alloying of Nb with Pd than with any other tested elements (Cu, Sn, Ni and Mo). However the alloying of Nb with Pd has not so great effect on the hydrogen solubility as alloying of V with Pd. For comparison, at 300 °, $S_{Nb-10Pd}/S_{Nb} \approx 0.30$ [11] while $S_{V-10Pd}/S_V \approx 0.13$ (Fig. 7).

Integrity of samples at hydrogen loading

The ribbon samples of all investigated alloys (from pure V to V-18.8 Pd) retained their integrity at hydrogenation at pressure up to 1 MPa, if temperature was higher than 300 °C. However, if the absorption experiment was carried out at lower temperature, all the investigated samples turned out broken into pieces after their extraction Particular temperature range where the destruction occurred for the V–Pd alloys with different content of Pd is indicated is Table 1. Our procedure of sample inspection (Sect. Observation of sample integrity) did not allow us determining the particular pressure (in the range from 10^{-4} MPa to 1 MPa) and content of solute hydrogen when the destruction occurred in the course of absorption experiment at given temperature.

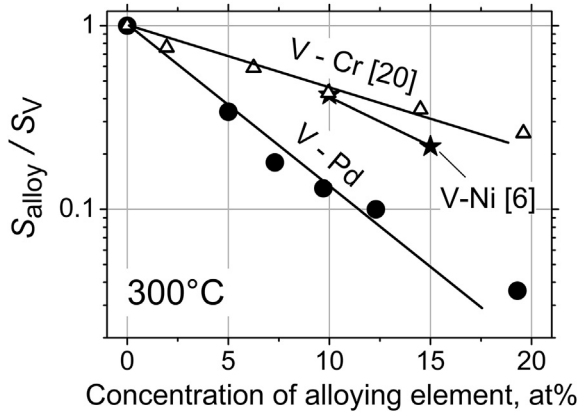


Fig. 7 – Capabilities of different alloying elements (Pd, Cr and Ni) to reduce hydrogen solubility in the substitutional alloys of V. The points for V–Pd alloys are found from *PCT* plot presented in Fig. 2, the points for V–Cr and V–Ni are computed from data by Lynch et al. [20] and Nishimura et al. [6] respectively.

Discussion

Possible mechanisms behind the effect of Pd on hydrogen solubility in substitutional V–Pd alloys

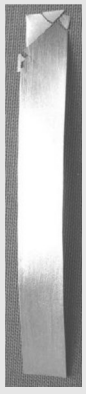
The main result of present work is observation of the great effect of alloying with Pd on the hydrogen absorption in V. What is the mechanism behind this phenomenon? How can it be explained that the substitution of relatively small portion of host atoms in the lattice leads to so radical reduction in hydrogen solubility?

The concept of antitrapping (or site blocking) is seemingly able to explain this phenomenon. In particular this approach was applied by Sakamoto et al. [17] to the explanation of significant reduction in the hydrogen solubility at the alloying of Pd with V (Fig. 6). The alloying atom simultaneously presents in several unit cells (the particular number depends on the crystal lattice [19]). Therefore even if the alloying atom blocks the absorption of H only in the nearest sites, the hydrogen solubility can be significantly reduced even at relatively small concentration of the alloying atoms. In this case the alloying results in decreasing hydrogen solubility due to reduction in the number of possible sites of sorption of H. Therefore the reduction in Siverts' constant, S, is expected to occur mainly via its pre-exponent (entropy) factor, S^*. The same cause is expected to lead to the reduction in the maximum hydrogen content of absorbed hydrogen, $(H/M)_{max}$, in substitutional alloys compared to $(H/M)_{max}$ in unalloyed metal ($(H/M)_{max} \approx 0.7$–0.8 in pure V, Fig. 2). On the other hand, the enthalpy of hydrogen solution is expected not to be noticeably changed because there are no alloying atoms in the nearest environment of sorption sites.

However, in spite of local non-uniformity, the substitutional V–Pd alloys behave as homogeneous with respect to the solution of H: the reduction in S with the growth of content of Pd occurs due to almost exclusively the increase of the enthalpy of solution while the pre-exponential factor, S^*, is nearly independent of content of Pd and close to S^* for pure V. As to the value of $(H/M)_{max}$, the maximum content of H absorbed in V–Pd alloys could not be reached experimentally in the investigated range of P and T, but the attained values of H/M for the V–Pd alloys were not very far from $(H/M)_{max}$ for pure V (Fig. 2), and that was obviously not the limit. Proceeding from that the values of $(H/M)_{max}$ for the V–Pd alloys are expected to be most probably rather close to that for pure V despite the drastic reduction in S.

It should be also noted that one of mechanisms that can be responsible for the above mentioned trapping/blocking phenomena is caused by an ability of alloying element either to

Table 1 – Photographs of specimens of different V-xPd alloys destructed in hydrogen media and temperatures of their destruction.

Content of Pd, x, at%	0	5	9.5	12.6	18.8
Temperature of destruction, °C	150	200–250	150	250–300	150–200
Photographs of specimens					

locally expand the host lattice, or vice versa, to contract it. According to this concept, the alloying elements contracting the host lattice should reduce S because of larger energy of elastic deformation required for the absorption of H atom in the vicinity of alloying atom. Vice versa, the elements expanding the lattice should increase S. This mechanism was found decisive for substitutional alloys of Pd [19] and in particular the great reduction in S due to the alloying of palladium with vanadium (Fig. 6) was interpreted in such manner [17]. On the basis of the 'expanded'–'contracted' classification of alloys one should expect an increase in S due to alloying of vanadium with palladium, however the opposite was observed according to the above. Taking into account this as well as the absence of any other evidences in favor of concept of local trapping/blocking, one can conclude that the corresponding mechanisms do not play any noticeable role in the case of substitutional V–Pd alloys. On the contrary, all the experimental data argue in favor of purely "global" (most probably through the change in the gross electronic structure) mechanism of the effect of Pd on the solution of H in V–Pd alloys.

The conclusion that substitutional atoms in the metal lattice do not block potentially available interstitial sites for hydrogen occupation was drawn also for substitutional V–Cr alloys [18,20]. Interestingly, Peterson and Nelson [18] have found that increasing of hydrogen solubility by the alloying of V with Nb and Ti is also a "global" (but not local/trapping) phenomenon. Further it was concluded from an overview of all the set of experimental data that the gross electronic structure is the most important factor in determining the enthalpy of solution of hydrogen, ΔH_{alloy}, in substitutional alloys of group IV–VI metals including vanadium [19].

One should note that regardless the specific mechanism being responsible for so strong dependence of enthalpy of solution, $\Delta H_{V\text{-}xPd}$, on palladium content, relative reduction in the binding energy of H atom with metal lattice, $E_{H\text{-}alloy}$, is far not as great as it might seem: totally less than 15% while $\Delta H_{V\text{-}xPd}$ changes more than 2 times (Fig. 4). That is because in general the enthalpy of hydrogen solution, ΔH_{alloy}, is the difference of two close quantities: $\Delta H_{alloy} = E_{H_2} - E_{H\text{-}aloy}$, where E_{H_2} is the binding energy of H atom in H_2 molecule ($E_{H_2} = 2.25$ eV/atom). Therefore ΔH_{alloy} is specifically sensitive to even small change in $E_{H\text{-}alloy}$ and the great change in ΔH_{alloy} by no means indicates any radical impact on the force or character of chemical interaction in the hydrogen–metal (alloy) system.

In connection with the influence of alloying on the binding energy one should also note that blocking effects of alloying atoms could be observed, if the absorption sites differed in $E_{H\text{-}alloy}$ by only a value of scale of RT (i.e. by ≈2%) depending on the presence or absence of Pd among the nearest metal atoms. Since the blocking effects were not observed, this difference in $E_{H\text{-}alloy}$ is even smaller. That seems to be rather surprising especially taking into account the great difference in atomic properties of Pd and V.

Suitability of V–Pd alloys as a membrane material

The large effect of Pd on the hydrogen solubility makes the V–Pd alloys rather promising material for selective membranes for hydrogen separation from gas mixtures. That allows using the alloys with a relatively small content of Pd. This gives two benefits. The first is the lower the concentration of alloying element the more ductile as a rule the alloy. Therefore the alloy with suitable hydrogen solubility is expected to be satisfactory rolled into the thin foils or drawn into the thin-walled tubes. Secondly the alloy may be not so expensive in spite of alloying with Pd.

If we compare the V–Pd and Nb–Pd alloys, the vanadium based alloys seem preferable. This is because (1) the effect of Pd on hydrogen solubility is substantially larger in vanadium based alloys (sect. The dependence on the concentration of Pd), and (2) the hydrogen solubility in pure V is substantially lower than that in pure Nb (by ≈2 times at 300 °C [14,15]). As a result the hydrogen solubility remains too large in Nb–Pd alloys even at relatively high content of Pd.

At the moment, V–Ni alloys are considered as the most promising binary alloys of V for membrane applications [10].

Despite higher cost of Pd, V–Pd alloys may be still preferable due to better ductility [7] and the low level of the alloying required for a proper reduction in the hydrogen solubility (Fig. 5).

The fact that the P–C behavior of absorbed hydrogen obeyed the Siverts' law in the range of non-dilute solutions (sect. Siverts' law: extending to the range of undiluted solutions) means that matrix-hydrogen interaction in the V–Pd alloys remains nearly constant in the range of rather high concentrations of absorbed hydrogen including the range of interest for membrane applications (up to H/M = 0.2–0.3) while the hydrogen–hydrogen interaction is significantly reduced due to the alloying.

Regardless the reasons of this phenomenon it allows additional increasing the operation pressure without exceeding the level of H/M permissible from the viewpoint of mechanical stability/durability. This is particularly important for membrane applications, when the maximum permissible concentration can be reached at the pressure significantly lower than the required one (at typical operating temperatures of 300 °C–400 °C). For example, when pure vanadium absorbs hydrogen at 350 °C, the value of equilibrium concentration H/M = 0.2 is achieved at the pressure as low as ≈0.017 MPa (Fig. 2(b)), while if Siverts' law were obeyed up to H/M = 0.2, the equilibrium pressure would be higher by more than 2 times. Thus alloying with Pd gives a double effect on the enhancement of operating pressure: it reduces the hydrogen solubility ($S_{V-Pd} < S_V$) and almost eliminates the deviation from Siverts' law. Due to this combined effect the maximum gain can be achieved in the regions of P and H/M that are most important for practice (P > 0.1 MPa and H/M between 0.1 and 0.2). For example, when V is substituted by V-9.7 Pd alloy at 350 °C, the equilibrium pressure increases by 65 times, if H/M = 0.2, and only by 30 times, if H/M < 0.02 (Fig. 2(b)).

One more important point is that both sides of composite membranes based on V and V-based alloys should be plated with Pd [1–7], and one of critical issues is the interdiffusion between the Pd coating and the bulk material. That results in a reduction of permeance in the course of operation due to the formation of V–Pd alloys [1,2,16]. The data presented in Fig. 6 can be considered in the context of this problem. They indicate that formation of V-xPd alloys in the range of $0 < x < 18.8$ at% (bcc solid solutions) hardly can result in reduction of global permeance of the composite membrane because hydrogen solubility in the V–Pd alloy remains not lower than that in the coating of pure Pd. And vice versa the formation of V-xPd alloys in the range of $90 < x < 100$ at% (fcc solid solutions) can reduce the permeance because the hydrogen solubility in the V–Pd alloy becomes much lower than that in Pd coating (Fig. 6). Besides the interdiffusion at V–Pd interface is expected to result first of all in formation of the Pd alloys doped with V due to preferable diffusion of refractory V in more fusible Pd. Of course, it would be desirable to know S_{V-xPd}/S_V in the whole range of x (including $18.8 < x < 90$ at%) for more comprehensive analysis of mechanisms behind the thermal degradation of composite membranes based on pure V and V–Pd alloys.

As to the mechanical stability, the free 100 μm-thick ribbons of all investigated alloys as well as of pure V demonstrated the maintenance of their integrity in hydrogen at pressure 1 MPa in temperature range 300 °C–400 °C that corresponds to typical conditions of membrane operation [1–11,16]. According to Yukawa et al. [3], the ductile-to-brittle transition in pure V occurs at the concentration H/M = 0.22 (this terminal concentration was found to be nearly independent of T [3]). Significantly higher concentration of hydrogen was absorbed in the samples of pure V and V–Pd alloys in our experiments (Fig. 2). Therefore the samples of pure V and most probably of V–Pd alloys were far beyond the zone of ductility at the hydrogenation. The maintenance of integrity in the zone of brittleness has been also observed with planar and tubular samples of pure V tested in permeation experiments [2].

One should also note that the reduction in temperature dependence of hydrogen solubility in V–Pd alloys in comparison with pure V (Figs. 3 and 5) is an additional favorable factor since the lower is temperature the greater reduction in hydrogen content relative to pure V and therefore the lower risk of mechanical destruction at membrane cooling in hydrogen media.

However, if temperature becomes lower a definite limit in hydrogen media, the destruction of all samples occurred (Table 1). It seems unlikely that this is caused by the formation of the hydride phase. There was no evidence of phase transition behavior (like plateau) in PCT curves for the investigated alloys at $T \geq 150$ °C (Fig. 2). Furthermore, our minimum temperature 150 °C is only slightly lower than the critical temperature of α–β phase transition for pure V. If the V–Pd alloys are capable of forming hydride phase, the critical temperature of its formation is most probably even lower than that for pure V. Whatever the reasons of the integrity failure, if the membrane is made of V–Pd alloy, the removal of hydrogen and dehydration of the membrane is necessary before its cooling down. That causes a definite inconvenience for practical use of the membranes based on the V–Pd alloys.

Conclusions

The alloying of V with Pd was found to greatly reduce the hydrogen solubility. Therefore Pd is rather promising as the alloying element for making V based alloy highly permeable to hydrogen but not so prone to dilation in hydrogen media as pure V. The higher the content of palladium in the alloy, the higher is the value of concentrations of absorbed hydrogen up to which the Siverts' law is obeyed. For the concentrations of Pd 9.7 at% or higher the Siverts' law extends far beyond the range of H/M where the solution can be considered as dilute (up to H/M ≈ 0.3). This phenomenon gives an additional gain for membrane applications of substitutional V–Pd alloys. The reduction in the hydrogen solubility at the alloying of V with Pd occurred only due to increase in the enthalpy of solution at nearly constant entropy factor. That is also a favorable factor lowering the risk of membrane destruction at its cooling in hydrogen media. Changes in the gross electronic structure of metal are most probably responsible for the effects of alloying on the hydrogen solubility in the substitutional V–Pd alloys.

Acknowledgments

We would like to thank Dr. V. Avakov for his support of this work as well as Mr. M. Kasatkin and Mr. I. Landgraph for their interest and help. We are also grateful to Mr. I. Grigoriady for technical assistance in the experiment. The authors acknowledge Saint-Petersburg State University for research grant 11.39.393. 2014.

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